Software Engineering

Lab 2: UML Use Case Diagrams

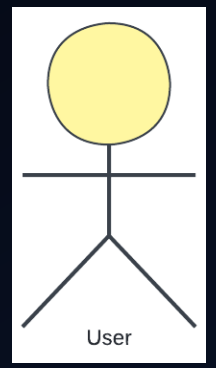
dr. inz Abdulla Juwaied

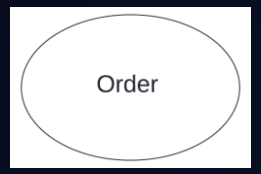
Learning Objectives

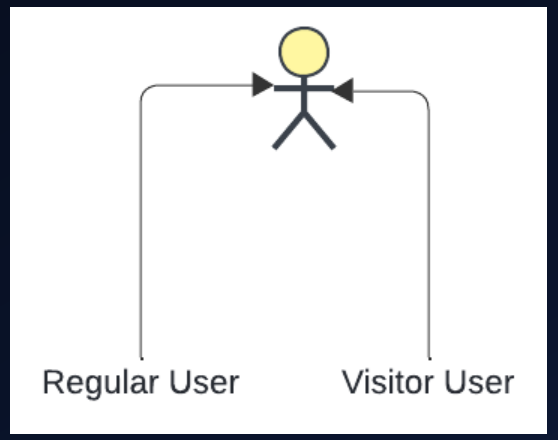
- Understand the core components of a use case diagram: actors, use cases, connectors, generalisation, and stereotypes (include/extend).
- Learn the rules and best practices for drawing clear and effective use case diagrams.
- Apply these concepts by creating a use case diagram for a given system.
- Practice writing use case descriptions linked to the diagram.

Identify the Core Components

- **Actor**: Represented by a stick figure. It can be a person, a group, or even another system.
- **Use Case:** Represented by an ellipse (oval). Inside this, you write the specific action or function (e.g., "Order").
- **Connector:** A simple line without arrows that connects an Actor to a Use Case.
- **Generalisation**: A line with a one-sided arrow used to divide an actor into more specific types (e.g., "User" divided into "Regular User" and “Visitor User").







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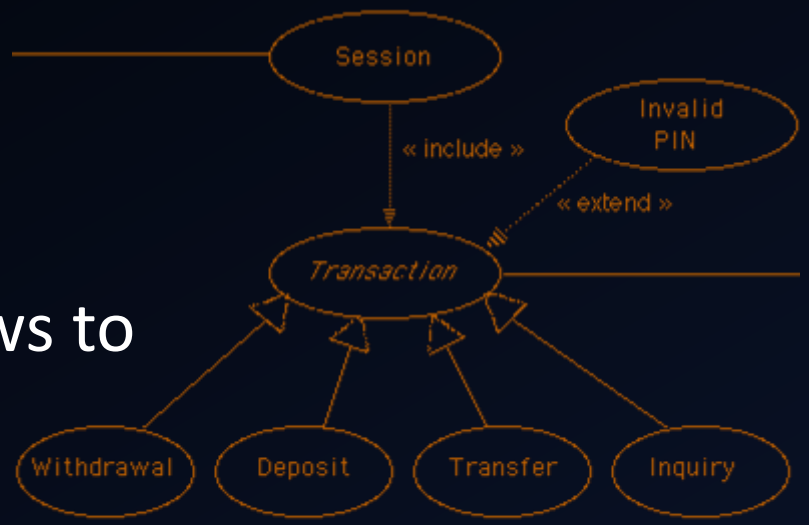


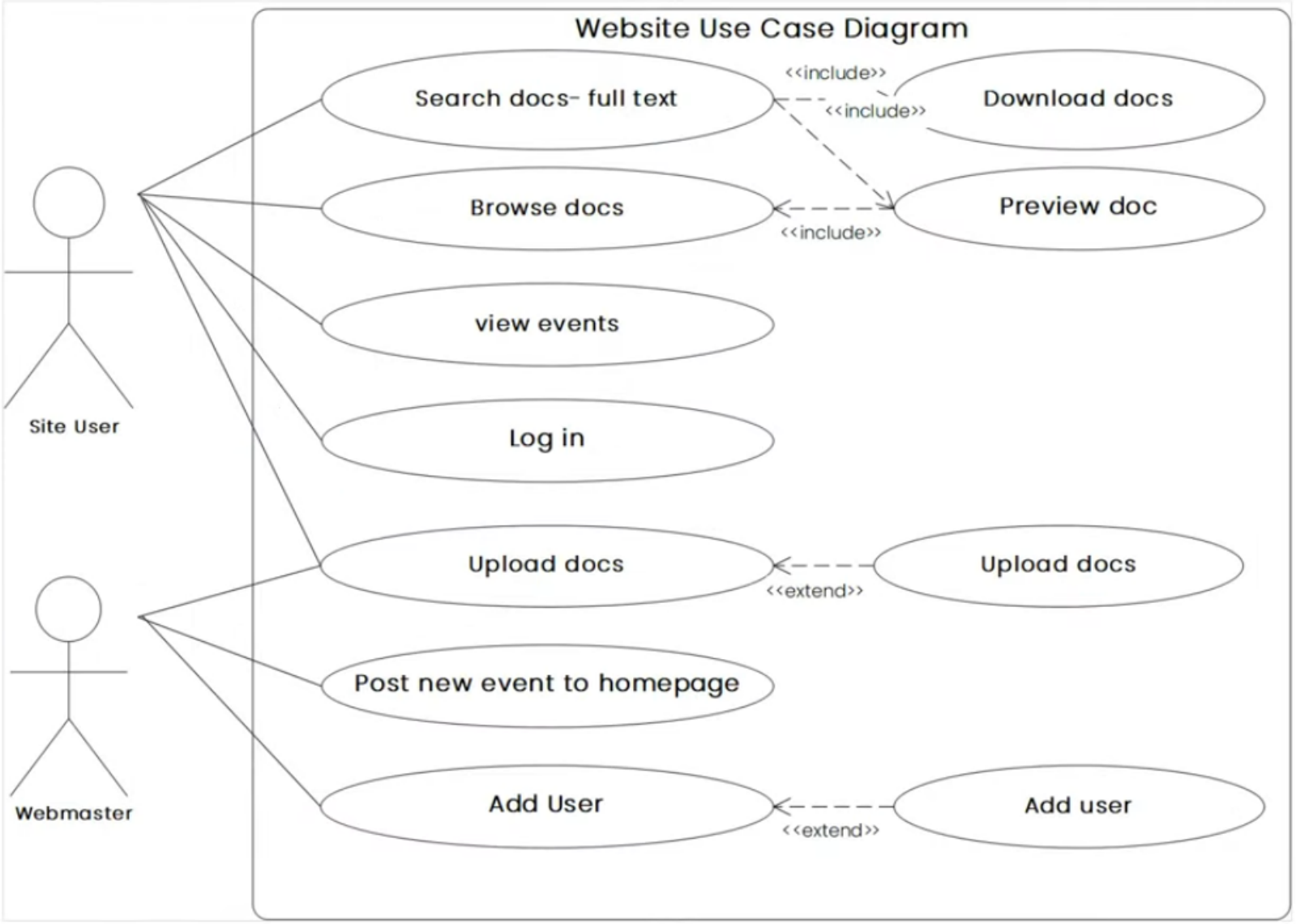
- **Relationships:** Represented by dotted arrows to show how use cases relate to each other:
  - **<<include>>**: Used for mandatory actions that *must* happen (e.g., you must "Login" to use a shopping app).
  - **<<extend>>**: Used for optional actions that *might* happen (e.g., giving "Feedback" after a purchase).

Dashed arrows with labels <<include>> (**mandatory**) and <<extend>> (optional).
Example: "Login" included in "Place Order"; "Provide Feedback" extends "Place Order".

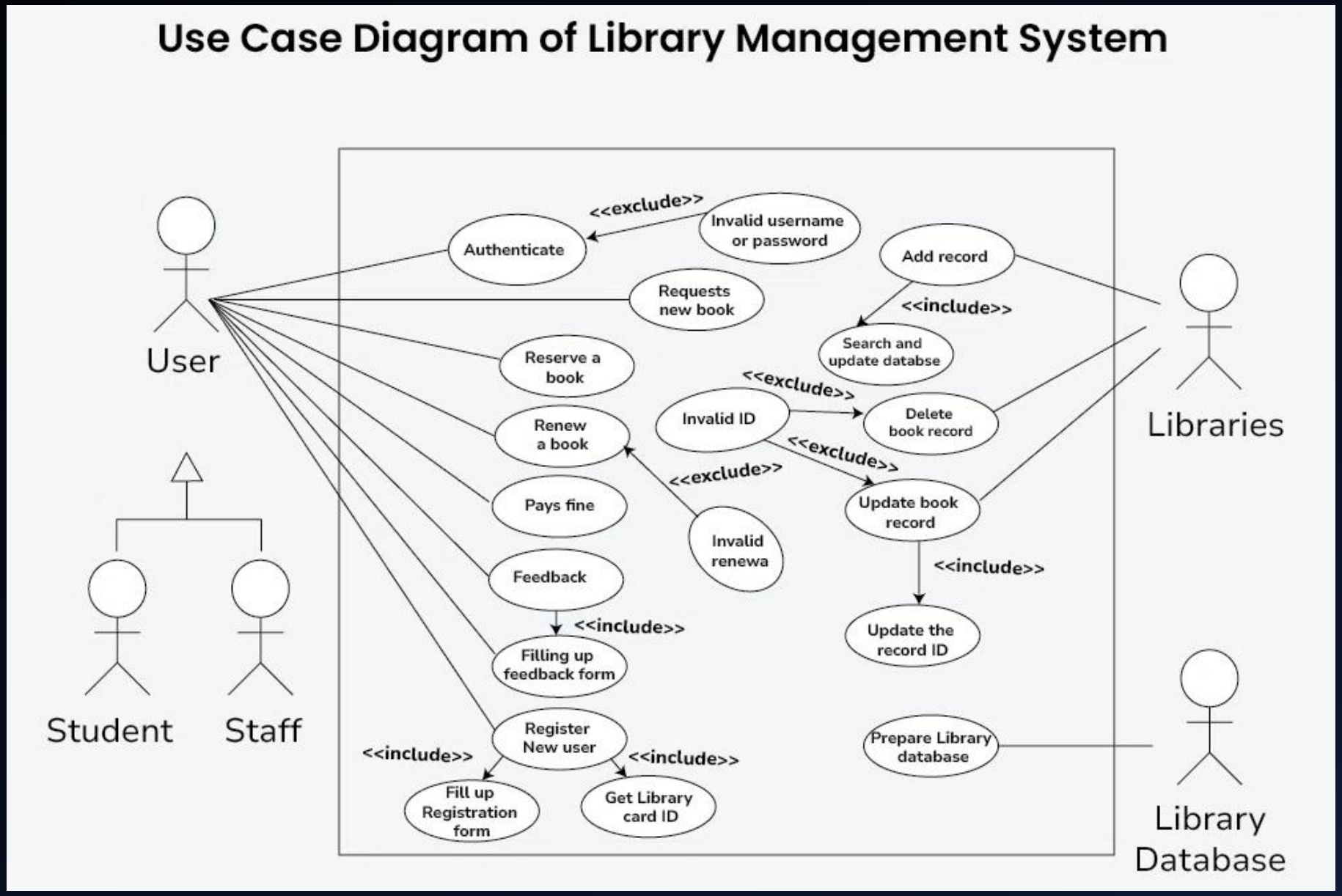
Follow the Essential Rules

- **Use Relationships:** Always try to include <<include>> and <<extend>> to show dependencies.
- **Keep it Simple:** Use Case names should be short and clear. For example, use "Order" instead of "Order Item in a Shopping App.
- **Actor Count:** Ensure you have at least two actors in your diagram for a complete system view.
- **Utilise Generalisation:** Use this concept to show specific categories of actors if applicable.
- **The System Boundary:** This is the most important final step. Draw a box around all your Use Cases, leaving the Actors **outside** the box. Name the box after your topic (e.g., "Online Shopping App")

Examples


Website Use Case Diagram
Search docs- full text
Download docs
<<include>>
<<include>>
Browse docs
Preview doc
<<include>>
view events
Log in
Site User
Upload docs
Upload docs
<<extend>>
Post new event to homepage
Add User
Add user
<<extend>>
Webmaster

Examples


Use Case Diagram of Library Management System
User
Authenticate
<<exclude>>
Invalid username or password
Requests new book
Reserve a book
Renew a book
Pays fine
Feedback
<<include>>
Filling up feedback form
Register New user
<<include>>
Fill up Registration form
<<include>>
Get Library card ID
Add record
<<include>>
Search and update databse
<<exclude>>
Invalid ID
Delete book record
<<exclude>>
Update book record
<<include>>
Update the record ID
<<exclude>>
Invalid renewa
Prepare Library database
Libraries
Library Database
Student
Staff

Task

- Choose a simple system (like an ATM, a Medical Clinic System, or Car insurance).
- List the primary and secondary actors.
- Define 3-5 key use cases.
- Determine if any use cases are mandatory (<<include>>) or optional (<<extend>>).
- Draw the diagram following the rules above, ensuring the system boundary is clearly labelled.
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- Remember to update your SRC Documents to include your topic with a Use Case diagram.